Project Synopsis

GTSRB Traffic-Sign Recognition

A complete, reproducible TensorFlow/Keras pipeline for the 43-class German Traffic Sign Recognition Benchmark.

Prepared for academic/project documentation
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1. Project Overview

This project develops an end-to-end image-classification system for recognizing German traffic signs. It covers data acquisition, integrity validation, exploratory data analysis, preprocessing, augmentation, model training, evaluation, command-line inference, a Colab workflow, and a presentation website.

2. Problem Statement

Traffic signs are visually diverse and may appear at different scales, viewpoints, lighting conditions, and backgrounds. The goal is to map a cropped traffic-sign image to one of 43 GTSRB class labels while preserving a reproducible evaluation boundary. This version addresses classification only; object detection and real-time camera streaming are outside scope.

3. Objectives

- Download and verify the complete public GTSRB classification archive.
- Profile class balance, image geometry, brightness, color, and representative examples.
- Train a compact CNN baseline and compare it with MobileNetV2 transfer learning.
- Use a grouped validation split, deterministic seeds, augmentation, checkpointing, and learning-rate control.
- Publish interpretable accuracy, macro F1, per-class metrics, confusion matrices, and learning curves.
- Provide local CLI commands, a Google Colab notebook, a static project website, and formal documentation.

4. Dataset and EDA Snapshot

Measure	Observed value
Training images	39,209
Official test images	12,630
Classes	43 (IDs 0–42)
Training class range	210–2,250 images
Archive MD5	7faafd0d78d831c7eb4d555 641a83a62
Source	Zenodo record 13 741 936

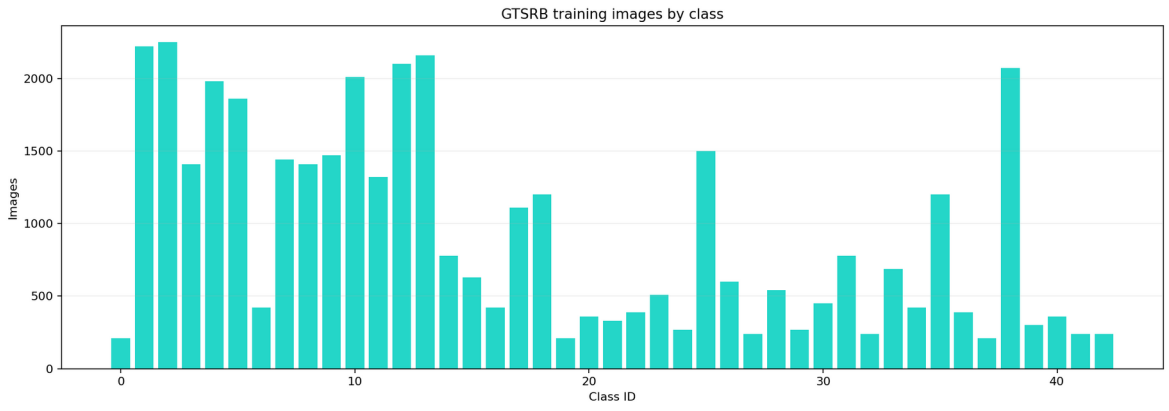


Figure 1. Training class distribution.

5. Methodology

Images are loaded from the official training directories and test CSV, resized to a consistent model-specific input size, normalized, and converted into TensorFlow datasets. Related image sequences are grouped before validation splitting to reduce near-duplicate leakage. Moderate random rotation, translation, zoom, brightness, and contrast transformations are applied during training.

Model	Role	Input
Custom CNN	Transparent baseline	64 × 64 × 3
MobileNetV2	Transfer-learning comparison	96 × 96 × 3

6. Demonstration Results

Model	Test accuracy	Macro F1
Custom Cnn	51.75%	36.44%
Mobilenet V2	53.09%	42.24%

The recorded runs use the full training and official test sets. The custom CNN was trained for three epochs and MobileNetV2 for one epoch as CPU-feasible demonstration runs; longer GPU training is recommended for a final benchmark.

7. Deliverables

Source modules live in src/traffic_sign_recognition. Reproducible commands are documented in README.md. The Colab notebook is notebooks/traffic_sign_recognition_colab.ipynb, the static website is site/dist, and the detailed report is docs/traffic_sign_recognition_project_report.docx.

8. Expected Impact

The project demonstrates how a machine-learning experiment can be made inspectable from raw data through inference. Its artifacts help students and reviewers connect data quality, model design, evaluation choices, and limitations in one coherent pipeline.